CSE226

Name – Himanshu rajawat

Registration no – 12317807

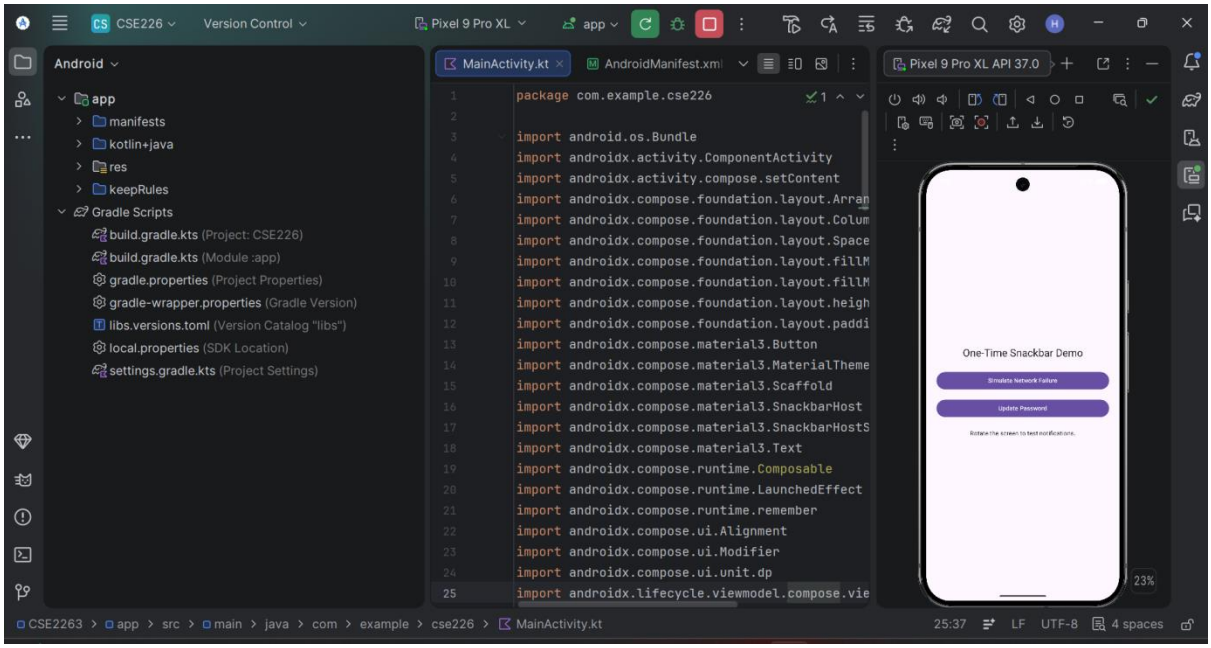
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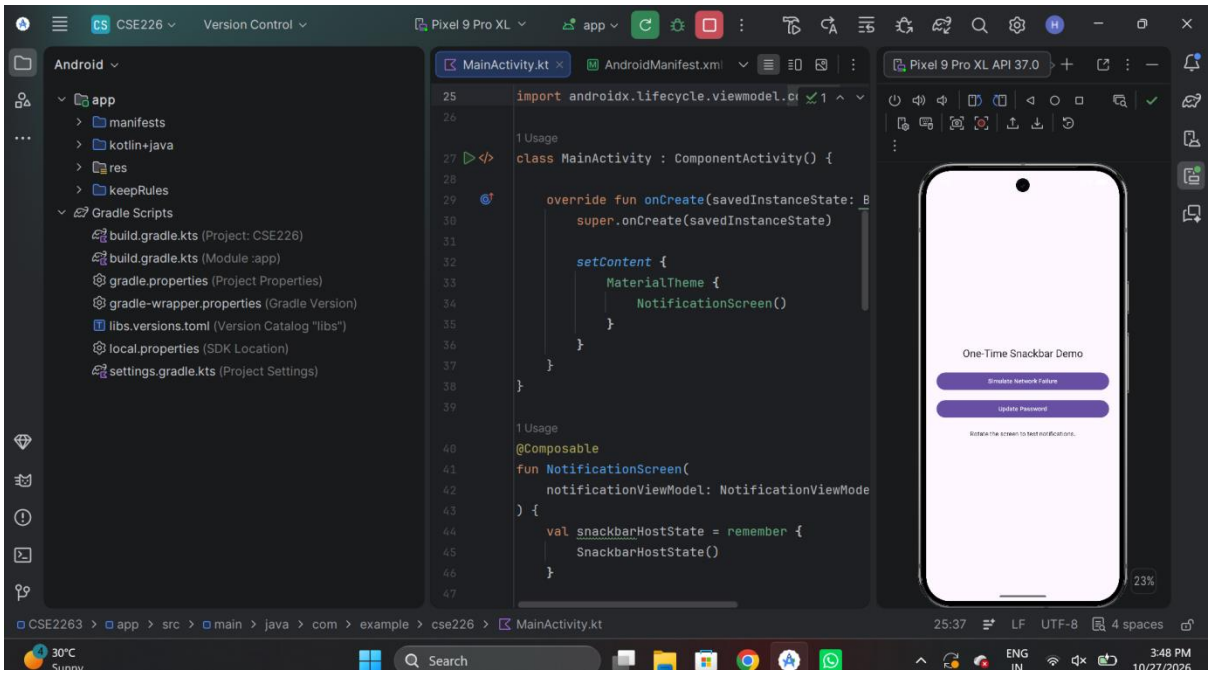
Github repo link: https://github.com/Himanshu629-netizan/CSE226-CA2

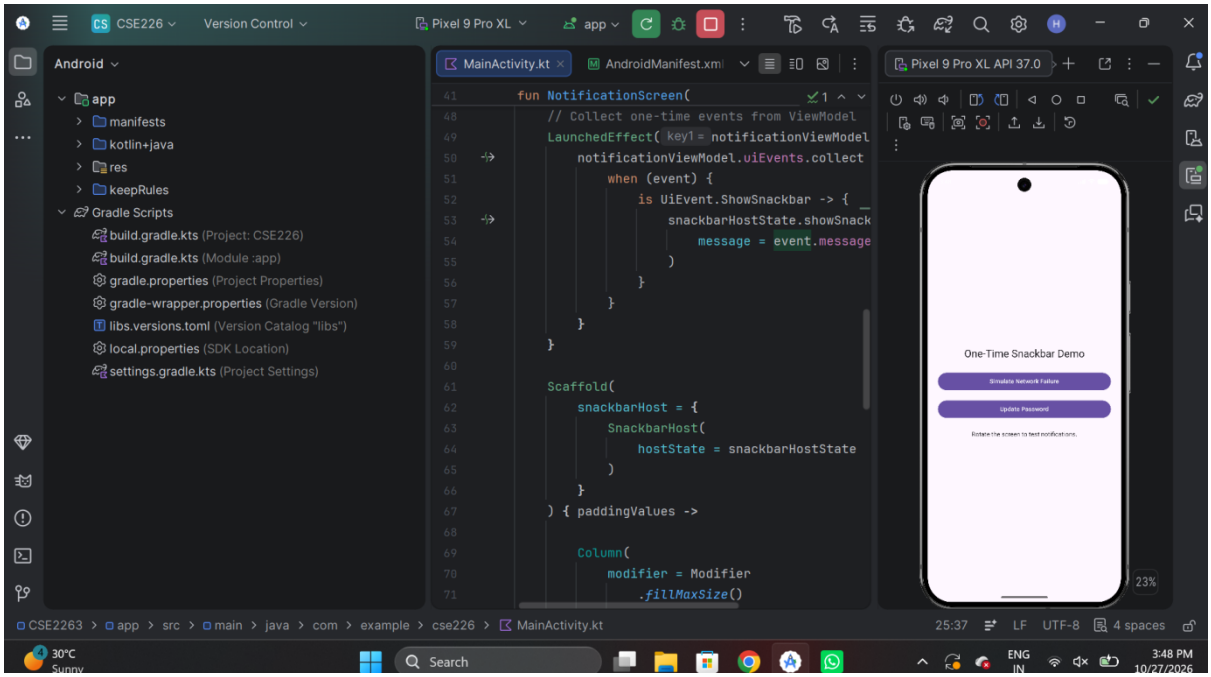
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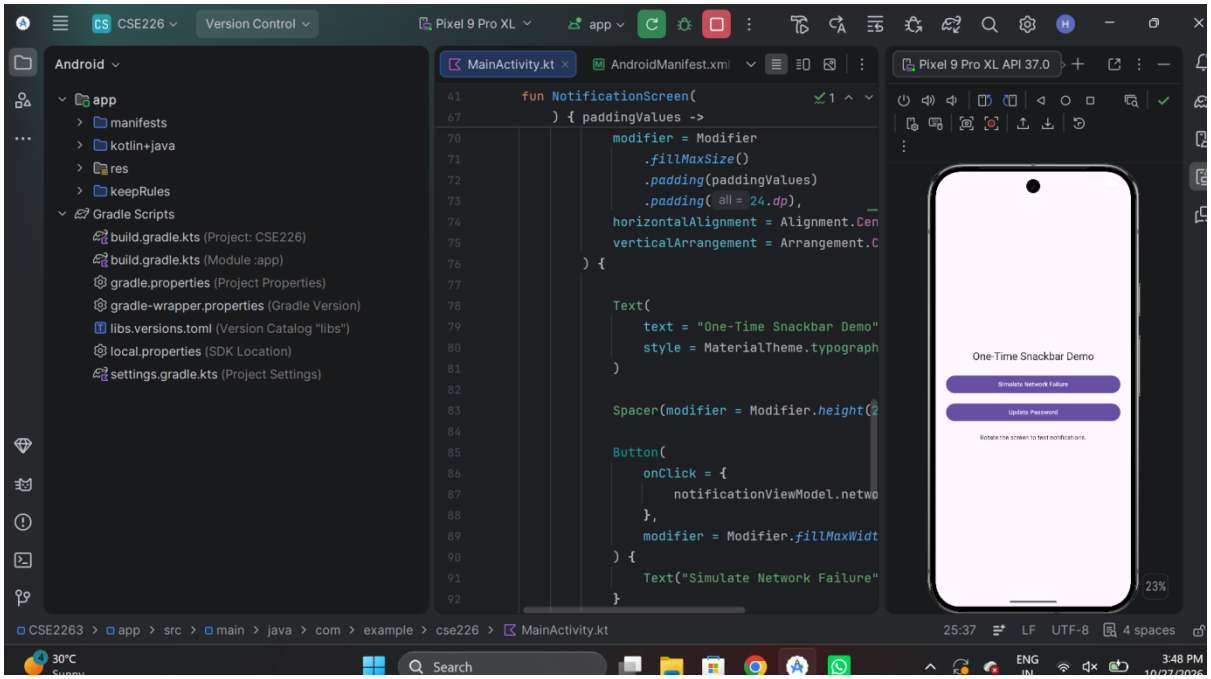
Set D
An enterprise messenger needs to broadcast transient notifications (e.g., "Network Connection Lost" or "Password Updated Successfully") to the user. These alerts must be displayed exactly once via a Snackbar and should not re-trigger when the user rotates their screen.
Model a one-time event pipeline using SharedFlow or Channel. Trigger an event from a ViewModel and collect it inside a LaunchedEffect block in the UI. Demonstrate that configuration changes (screen rotations) do not cause the notification event to re-emit or reshow on the UI.

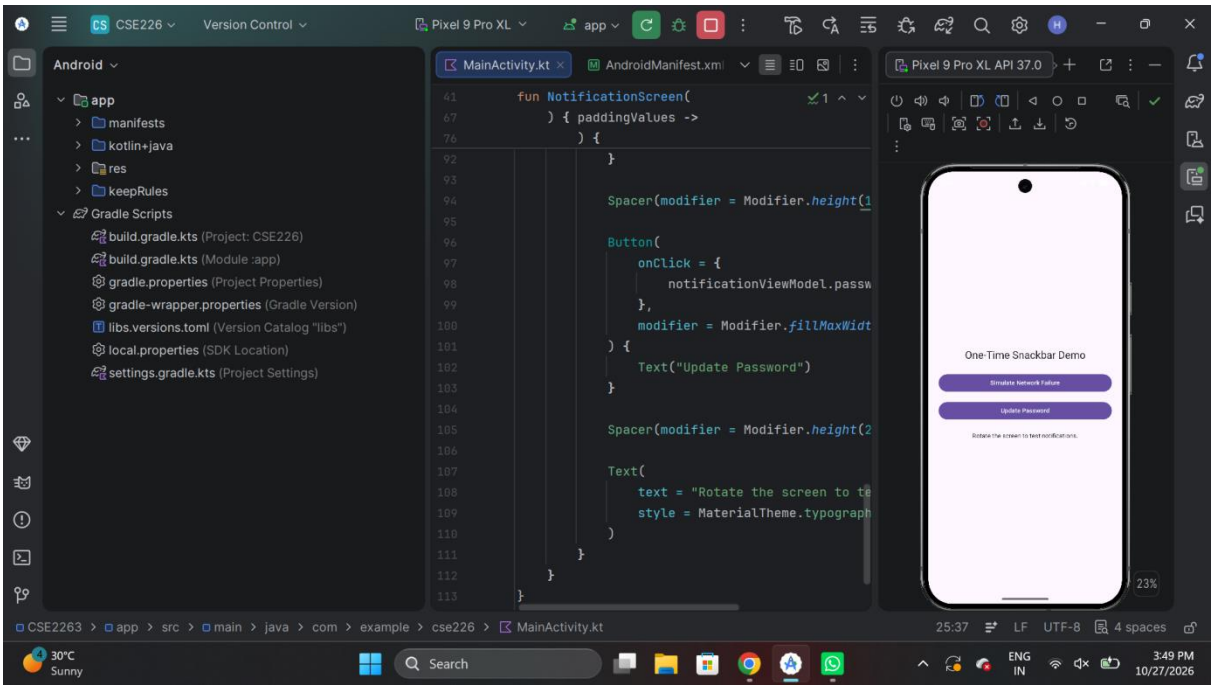
Here is my complete code of execution.


CSE226
Version Control
Pixel 9 Pro XL
app
Android
app
manifests
kotlin+java
res
keepRules
Gradle Scripts
build.gradle.kts (Project: CSE226)
build.gradle.kts (Module :app)
gradle.properties (Project Properties)
gradle-wrapper.properties (Gradle Version)
libs.versions.toml (Version Catalog "libs")
local.properties (SDK Location)
settings.gradle.kts (Project Settings)
MainActivity.kt
AndroidManifest.xml
package com.example.cse226
import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.Arran
import androidx.compose.foundation.layout.Colum
import androidx.compose.foundation.layout.Space
import androidx.compose.foundation.layout.fillM
import androidx.compose.foundation.layout.fillM
import androidx.compose.foundation.layout.heigh
import androidx.compose.foundation.layout.paddi
import androidx.compose.material3.Button
import androidx.compose.material3.MaterialTheme
import androidx.compose.material3.Scaffold
import androidx.compose.material3.SnackbarHost
import androidx.compose.material3.SnackbarHostS
import androidx.compose.material3.Text
import androidx.compose.runtime.Composable
import androidx.compose.runtime.LaunchedEffect
import androidx.compose.runtime.remember
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.unit.dp
import androidx.lifecycle.viewmodel.compose.vie
Pixel 9 Pro XL API 37.0
One-Time Snackbar Demo
Simulate Network Failure
Update Password
Rotate the screen to test notifications.
23%
CSE2263 > app > src > main > java > com > example > cse226 > MainActivity.kt
25:37
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UTF-8
4 spaces


CSE226
Version Control
Pixel 9 Pro XL
app
Android
app
manifests
kotlin+java
res
keepRules
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build.gradle.kts (Module :app)
gradle.properties (Project Properties)
gradle-wrapper.properties (Gradle Version)
libs.versions.toml (Version Catalog "libs")
local.properties (SDK Location)
settings.gradle.kts (Project Settings)
MainActivity.kt
AndroidManifest.xml
import androidx.lifecycle.viewmodel.c
1 Usage
class MainActivity : ComponentActivity() {
override fun onCreate(savedInstanceState: B
super.onCreate(savedInstanceState)
setContent {
MaterialTheme {
NotificationScreen()
}
}
}
}
1 Usage
@Composable
fun NotificationScreen(
notificationViewModel: NotificationViewMode
) {
val snackbarHostState = remember {
SnackbarHostState()
}
Pixel 9 Pro XL API 37.0
One-Time Snackbar Demo
Simulate Network Failure
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Rotate the screen to test notifications.
23%
CSE2263 > app > src > main > java > com > example > cse226 > MainActivity.kt
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Search
ENG
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3:48 PM
10/27/2026


CSE226
Version Control
Pixel 9 Pro XL
app
Android
MainActivity.kt
AndroidManifest.xml
Pixel 9 Pro XL API 37.0
app
manifests
kotlin+java
res
keepRules
Gradle Scripts
build.gradle.kts (Project: CSE226)
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gradle.properties (Project Properties)
gradle-wrapper.properties (Gradle Version)
libs.versions.toml (Version Catalog "libs")
local.properties (SDK Location)
settings.gradle.kts (Project Settings)
fun NotificationScreen(
// Collect one-time events from ViewModel
LaunchedEffect(key1 = notificationViewModel
notificationViewModel.uiEvents.collect
when (event) {
is UiEvent.ShowSnackbar -> {
snackbarHostState.showSnack
message = event.message
)
}
}
}
}
Scaffold(
snackbarHost = {
SnackbarHost(
hostState = snackbarHostState
)
}
) { paddingValues ->
Column(
modifier = Modifier
.fillMaxSize()
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23%
CSE2263 › app › src › main › java › com › example › cse226 › MainActivity.kt
25:37
LF
UTF-8
4 spaces
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Search
ENG
IN
3:48 PM
10/27/2026


CSE226
Version Control
Pixel 9 Pro XL
app
Android
MainActivity.kt
AndroidManifest.xml
Pixel 9 Pro XL API 37.0
app
manifests
kotlin+java
res
keepRules
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build.gradle.kts (Module :app)
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gradle-wrapper.properties (Gradle Version)
libs.versions.toml (Version Catalog "libs")
local.properties (SDK Location)
settings.gradle.kts (Project Settings)
fun NotificationScreen(
) { paddingValues ->
modifier = Modifier
.fillMaxSize()
.padding(paddingValues)
.padding(all = 24.dp),
horizontalAlignment = Alignment.Cen
verticalArrangement = Arrangement.C
) {
Text(
text = "One-Time Snackbar Demo"
style = MaterialTheme.typograph
)
Spacer(modifier = Modifier.height(2
Button(
onClick = {
notificationViewModel.netwo
},
modifier = Modifier.fillMaxWidt
) {
Text("Simulate Network Failure"
}
One-Time Snackbar Demo
Simulate Network Failure
Update Password
Rotate the screen to test notifications.
23%
CSE2263 › app › src › main › java › com › example › cse226 › MainActivity.kt
25:37
LF
UTF-8
4 spaces
30°C
Sunny
Search
ENG
IN
3:48 PM
10/27/2026


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app
manifests
kotlin+java
res
keepRules
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MainActivity.kt
AndroidManifest.xml
fun NotificationScreen(
) { paddingValues ->
Spacer(modifier = Modifier.height(1
Button(
onClick = {
notificationViewModel.passw
},
modifier = Modifier.fillMaxWidt
) {
Text("Update Password")
Spacer(modifier = Modifier.height(2
Text(
text = "Rotate the screen to te
style = MaterialTheme.typograph
Pixel 9 Pro XL API 37.0
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Simulate Network Failure
Update Password
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